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Applying machine learning techniques in weather forecast modeling

Michał Ciszynski^a, Kornel Chrominski^{a,*}

^aUniversity of Silesia in Katowice, Institute of Computer Science, Bedzinska 39, 41-200 Sosnowiec, Poland

Abstract

Following work concerns analysis of use of artificial intelligence in meteorological applications for weather forecasting problems. In order to interpret effect of appropriate models, dataset ERA5 created by European Centre for Medium-Range Weather Forecasts was utilized. This datasource contains data from global measurements assimilated by newest numerical models. Analysis was focused on short term weather forecast in time range of 24 hours, and geographical range of one location. Created models were optimized by hiperparameter tuning. Among considered models were SVR, logistic regression, SGD, K-nearest neighbors, gaussian regression, decision tree, random forest.

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1. Introduction

This article is focused on the analysis and evaluation of the application of machine learning models for weather forecasting and weather modeling purposes. Particular emphasis was placed on the comparison of selected models and the analysis of the results due to possible improvements. In addition, the effectiveness of AI algorithms was compared due to the variables analyzed. Since not all analyzed algorithms proved to be suitable for the task at hand, the drawbacks of selected algorithms were also analyzed.

Nomenclature

ML - machine learning

NWP- numerical weather prediction

AI - Artificial intelligence

* Corresponding author. Tel.: +48-32-368-97-63

E-mail address: kornel.chrominski@us.edu.pl

MSE - mean square error

MAE- absolute error

1.1. Background and motivation

Weather prediction is very important nowadays, not only in terms of information for people to help with planning. Predicting the weather forecast is also very important for the economy and agriculture. Monitoring and predicting weather is used in areas such as agro-culture, controlling the state of air pollution, predicting road conditions to improve road safety, predicting the spread of natural fires, predicting natural disasters, applying renewable energy sources, and many others. Accurate prediction of weather conditions is critical to the proper functioning of many organizations. The U.S. Environmental Information Centers estimated the amount of damage resulting from weather in 2015 at \$7.9 billion. On the other hand, solar forecasting capabilities offer a chance to save \$455 million by 2040 in the use of renewable energy sources [1].

The first attempts at “handwritten” NWP models were initiated by Lewis Fry Richardson in 1922 and were inspired by the idea of predicting the weather using formulated laws of physics. The formulation of nonlinear differential equations describing the weather created an initial value problem based on current weather measurements. Interactions from thermodynamics, dynamics, chemical processes, as well as biological processes must be taken into account to achieve the desired goal. The first computer-aided weather forecasts were created as early as 1950, but it was not until 1970 with the development of supercomputers that solving the full set of equations was possible [5].

The basic equations describing the behavior of weather are the Navier-Stokes equations, the equations of conservation of mass, the first law of thermodynamics and Clapeyron’s equation. With the help of these, the state of wind, pressure, density, temperature can be modeled, describing the state of the atmosphere. Due to the lack of an analytical solution to these equations, a numerical solution is a necessity. Physical processes occurring on a scale too small to be included when discretizing the space must be described using parameterizations that attempt to approximate their effect on the overall solution. In many cases, additional simplifications and rounding are taken into account to simplify the solution process. Some global models and almost all local models use the finite difference method in all degrees of freedom of the system, and some global models use the finite difference method in the vertical direction and spectral methods in the direction parallel to the earth’s surface [2][3][4].

Due to the chaotic characteristics of the equations describing the weather, small changes in the initial conditions can result in outliers. Ensemble models have been developed to describe and predict weather forecast error. The nonlinear complexity of the system means that approaches to describing forecast accuracy based on purely statistical methods do not produce the desired results. Therefore, a collection of multiple models initialized with slightly different initial conditions is used to test how much the results obtained can vary. Early methods of specifying initial conditions were based on analysis of graphical weather charts. Subsequently, various data interpolation methods were replaced by data assimilation algorithms based on optimal control theory. The calculation of actual values describing the atmosphere can be described as inferred Bayesian based on the results of measurements and their uncertainties. These calculations focused on minimizing the objective function are carried out in a four-dimensional space to obtain a physically possible result evenly distributed over three spatial dimensions and time. In practice, the use of NWP models for weather prediction purposes requires the assimilation of a large amount of data from different sources.

The method used today for data input is 4D-Var assimilation (used since 1980). In this algorithm, an observation operator is created for each type of observation, allowing the internal parameters of the NWP model to refer to a particular type of observation. During the data assimilation process, the model’s internal parameters are adjusted to best reflect actual weather conditions. In many cases, the computational complexity of NWP requires a reduction in the volume of data derived from high-density satellite observations. The resolution of data derived from satellite readings far exceeds the capabilities of previously used models created by ECMWF that have a horizontal resolution of 9 km. Moreover, in the case of ensemble models to reduce computational costs, the resolution is reduced to 18 km. Processes that cannot be faithfully reflected by simulation due to the insufficient resolution of the NWP are subject to parameterization. Such processes include cloud formation, the amount of sunlight reaching the ground, the aerodynamic drag of mountain peaks, or the formation of water droplets. The final result of the conducted NWP

simulation must be further enriched by applying model output statistics (MOS). As part of this, the predicted state of the model described by complex parameters is converted into quantities easily interpreted by humans.

Nowadays, it is increasingly apparent that existing models for weather forecasting are proving less and less successful. With the importance of predicting the weather forecast itself, developing new ways to predict the weather seems to be very important. With the current development of artificial intelligence methods, it is worth looking into machine learning techniques for weather prediction [6]. The biggest problem in developing AI models for weather prediction is maintaining physical constraints that are not directly part of the model, unlike NWP algorithms. Moreover, the way neural networks analyze weather data is impossible to interpret, although there are methods that attempt to visualize what features in the input data are interpreted by the neural network, in many cases, NNs can be treated as a “black box” that transforms the input data into the desired results. Difficulties associated with the use of machine learning for weather forecasting also include the complex statistics of the data being analyzed, multiple sources of data noise, and the high dimensionality of the data describing the weather [7].

2. Machine learning methods

The problem of weather forecasting can be described using the regression method. It involves statistically modeling the response of unknown function values based on known values of independent variables. In weather prediction, known values of atmospheric measurements can be taken as independent variables and predicted values can be estimated using regression. Classification methods can also be useful for detecting and predicting weather phenomena. In this article, the following machine learning regression methods will be used for weather forecasting [8] [9] [10]:

- SVR - is a regression method that uses a support vector machine to create predictions. It is based on the creation of a hyperplane aimed at separating observations from the training set as accurately as possible. The determined hyperplane equation is then used to classify and calculate regressions for scientific data. The calculation of the hyperplane equation boils down to the optimization problem of maximizing the distance of the hyperplane from the points belonging to each class. In addition to carrying out linear regression, SVR can create nonlinear regressions by applying a “kernel trick,” implicitly mapping the input data domain to a multidimensional space with different characteristics.
- Logistic Regression - is a statistical model that predicts the occurrence of an event based on the calculation of a logit function from a linear combination of independent variables.
- SGD - is a method for training machine learning models through iterative optimization of the objective function. The application of SGD has obtained good and natural language processing. Under this algorithm, the starting point selected by the user is subjected to iterative optimization by selecting the direction of the largest decrease in the error function. In this way, the solution space is searched in a very efficient way. Unfortunately, ensuring a global minimum is not always guaranteed.
- KNN - is a simple model that can be used for both regression and classification. It involves selecting k nearest neighbors from a training set to determine the value for a new observation. After selecting the most similar observations from the training set, the value of the objective function is averaged.
- Gaussian regression - is a non-parametric form of the Bayesian method for making predictions. Instead of calculating the distribution of function parameters, this method can be used to calculate the distribution of the objective function.
- PLS - Partial least squares regression- is an algorithm that uses the decomposition of data due to principal component analysis (PCA) to use the least squares method in the next regression.
- Decision Tree-is a tree-structured model that aims to predict the value of a goal function based on a subset of observation attribute values. Decision trees are built based on recursive partitioning of training data due to the attribute optimizing measure of tree fit (Gini, information gain). Due to the different measures used and methods of building the tree, ID, C4.5, C5.0 and CART algorithms are distinguished.

3. Experiments

This section presents the results of research on the effectiveness of using machine learning methods in weather prediction.

3.1. Research methodology

The study used the ERA5 meteorological dataset collected by the European Center for Medium-Range Weather Forecasts (ECMWF) containing hourly data on atmospheric, land, and oceanic variables. For the study, the data was limited to values describing wind speed, temperature, solar radiation, pressure, precipitation, evaporation, and changes in watercourse levels.

Although ERA5 contains information dating back to 1940 and covering the entire earth with a 30-km resolution grid, for the study it was limited to a single location from the time interval 2003 - 2023 and contains only observations from a single month - January. The dataset used is a reanalysis dataset - it is based on historical data augmented with advanced modeling methods and data assimilation algorithms. As a comparison of ML algorithms, the following work uses linear regression algorithm, random forest, decision tree, PLS regression, Gaussian process regression, k nearest neighbors, SGD regression, and support vector regression. In addition, in an attempt to obtain the best possible results, the optimization of hyperparameters was carried out using an exhaustive search of the space of available parameters. To speed up this process, the quality of hyperparameter evaluation is increased as the algorithm runs and better results are found.

3.2. Data set

The data used in this paper was downloaded from the open source Copernicus Climate Change Service. This site provides a publicly accessible API through which, any user is able to send a request for the range of data they are interested in. With this API, it is possible to specify the time frame, geographic range and types of variables. Selected variables:

- Vertical and horizontal components of wind speed (u10, v10);
- Temperature 2 meters above ground level (t2m);
- Changes in watercourse levels (ro);
- Solar radiation (ssr);
- Thermal radiation (str);
- Pressure (sp);
- Total evaporation (e);
- Precipitation (tp).

The data consists of the 9 previously mentioned parameters and 12,648 hourly observations collected from a 20-month interval. All data are from January and were taken from the 20-year range 2003 - 2023. A location in Poland was chosen as the location.

All tables should be numbered with Arabic numerals. Every table should have a caption. Headings should be placed above tables, left justified. Only horizontal lines should be used within a table, to distinguish the column headings from the body of the table, and immediately above and below the table. Tables must be embedded into the text and not supplied separately. Below is an example which the authors may find useful.

Figure 1 presents autocorrelation plots of the selected dataset over 4 days. The occurrence of daily cycles and their influence on individual measurements can be clearly seen. An increase in autocorrelation values for a shift that is a multiple of a day is evident for temperature, evaporation, solar radiation, and thermal radiation. The rest of the variables show a steady decrease in correlation with increasing lag. Particularly low values can be observed for precipitation, for which the autocorrelation quickly falls below 0.2, indicating the extremely chaotic nature of the phenomenon.

Table 1. Dataset statistics.

variable	mean	min	max	std
u10 [m/s]	1.11	-6.94	10.15	2.64
v10 [m/s]	1.28	-9.02	10.70	2.77
t2m [K]	271.40	247.80	285.70	5.80
e [m]	-1.16E-5	-1.88E-4	4.78E-5	1.78E-5
ro [m]	2E-5	3E-6	3.74E-4	1.4E-5
ssr [J/m2]	8.51E4	-3.12E-2	1.05E6	1.66E5
str [J/m2]	-1.28E5	-4.17E5	3.20E4	8.64E4
sp [Pa]	9.73E4	9.33E4	9.99E4	995
tp [m]	9.77E-5	0	2.51E-3	1.95E-4

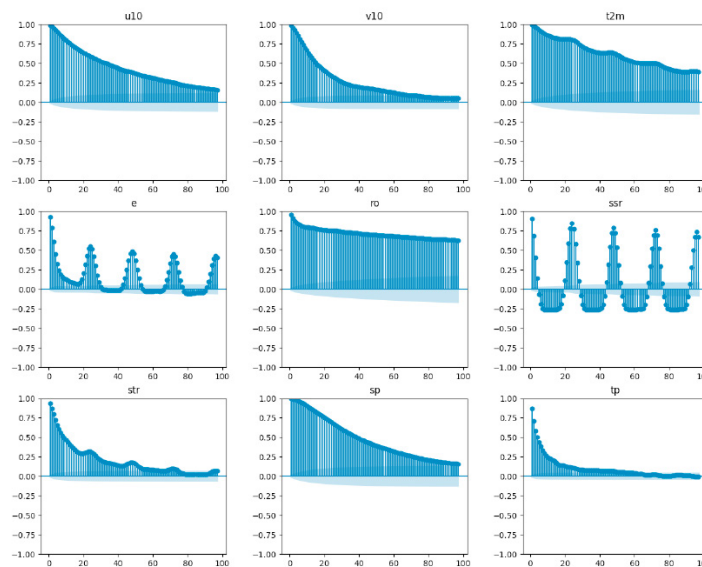


Fig. 1. Autocorrelation of selected variables against a lag of 1 hour to 96 hours

3.3. Results

The following section compares the results obtained for the models considered and performs an analysis due to the accuracy of the generated predictions. The analysis was carried out due to the overall value of accuracy, but also taking into account the characteristics of individual variables.

As can be seen in Fig. 2 a, there was a decrease in accuracy for each algorithm relative to the test data. In the case of the KNN model and Gaussian regression, the negligible error in the training data is because both models are based on learning by example and their evaluation of training data is unreliable.

Random forest turns out to be an algorithm that can fit the training data very well together with maintaining generality for the rest of the data, achieving good results on the test set. The absolute error values reach higher values than the MSE (Fig. 2) b. Moreover, the discrepancy between the error for the training set and the test set using the MAE metric becomes smaller. The overall distribution of error relative to the models seems to be similar.

For all models, the correlation coefficient between actual and predicted values is above 0.9 (see Fig. 3). This indicates a good fit of the models to the predicted data, although it is worth pointing out that the autocorrelation coefficient for many parameters remained above 0.8 for delay values exceeding a couple of hours.

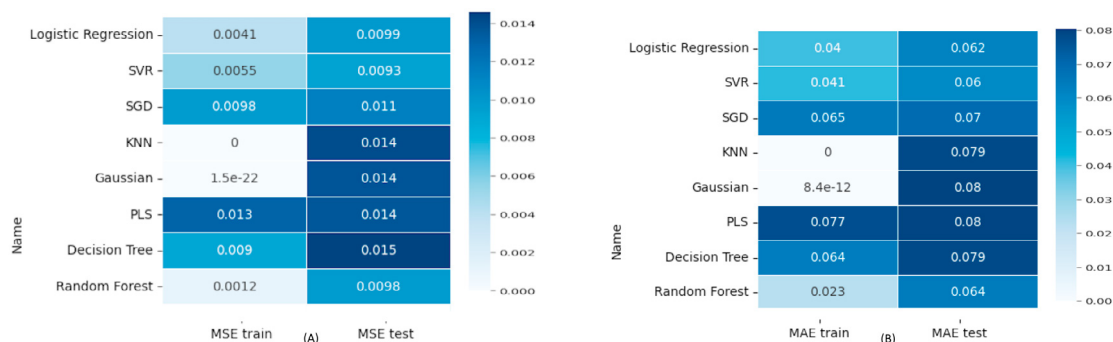


Fig. 2. (A) Mean square error for the created models, (B) Average absolute error for the created models

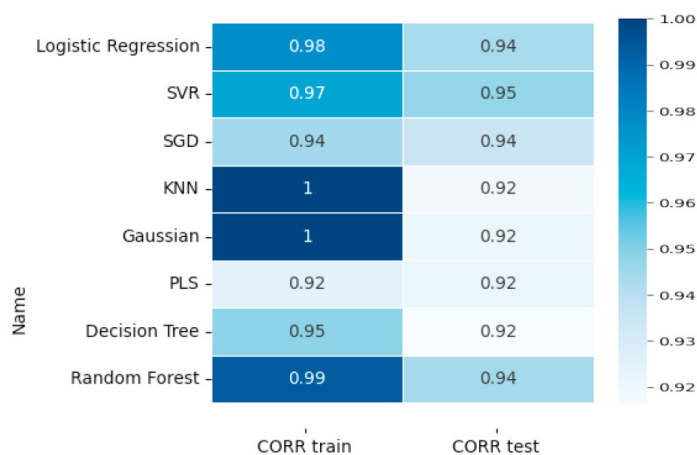


Fig. 3. Ranking of the created models due to the mean square error

Figure 4 shows the ranking created due to the value of MSE. The best algorithm in this regard was SVR. The discrepancy between the best model and the worst was 0.0057. The algorithms achieving the worst results were KNN and decision trees.

3.4. Construction of references

To further approximate the behavior of the various models, summaries of real-time runs and those generated by machine learning algorithms were created. While the metrics presented give a preliminary understanding of the degree of fit of the models and their accuracy, graphical analysis is key to understanding which phenomena are predicted well and which pose problems. The waveforms presented here show weekly waveforms consisting of seven 24-hour forecasts.

One of the best-evaluated algorithms was support vector regression, a summary of the time series for the SVR model is shown in Figure 5. In this case, we see good behavior against cyclic waveforms such as solar radiation and evaporation. In terms of thermal radiation, a good representation of the mean value of the time series is evident, but a problem with matching individual spikes in values. The sudden jump in precipitation values was completely ignored in the forecast and indicates problems in predicting sudden events with large amplitudes. Both wind speed and temperature values were faithfully reproduced by the model created, and the following graphs show how the lines of the predicted values follow the predicted values. In the case of atmospheric pressure, after a sudden drop, the model was able to adjust again to the measured data and continue to follow the created trend.

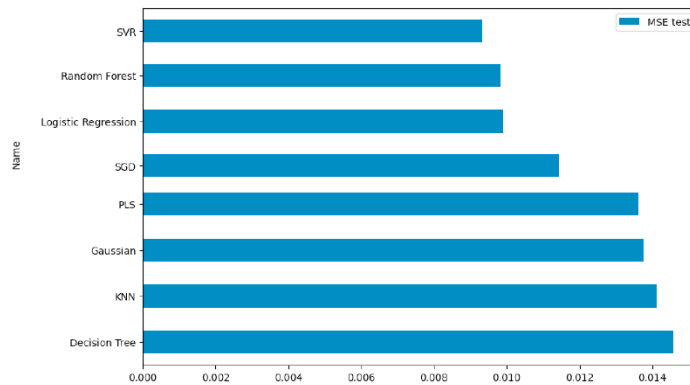


Fig. 4. Correlation coefficient for the created models



Fig. 5. Time courses for the SVR algorithm. The yellow line shows actual data and the blue shows predictions generated by the model

The ability of individual models to regress against particular attributes was also analyzed. Thus, for example, On figure 6 are presented the mean squared error for the SVR model, which was one of the best-performing models. The graph shows that the largest error is generated by forecasting thermal radiation. Problems in regression against this attribute may be due to its high variance and dependence on physical processes not included in the dataset, among other reasons.

When comparing models, the same metrics can be used as for evaluating prediction against real data. The highest correlation values are seen between SVR, logistic regression, random forest models (see Fig. 7). These are the models that obtained the best results in the experiments conducted. Another group of algorithms close behind is Gaussian regression, PLS, which, although they did not get the best results, was among the algorithms with average performance. The decision tree exemplifying the worst of the algorithms under consideration is at a distance from the other models.

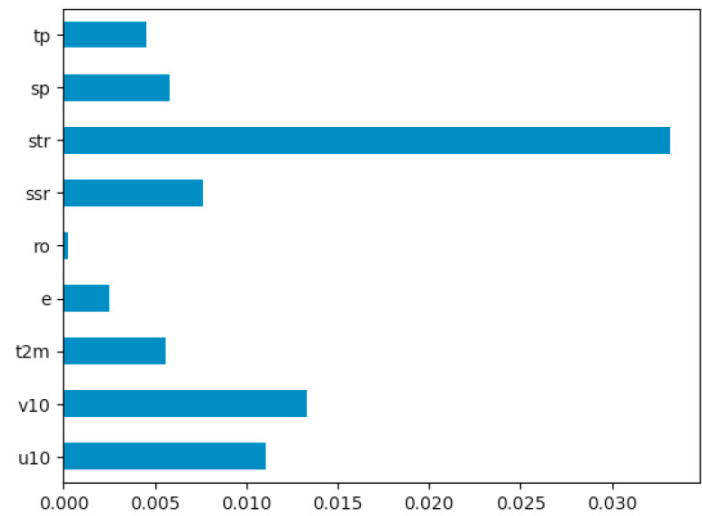


Fig. 6. Time courses for the SVR algorithm. The yellow line shows actual data and the blue shows predictions generated by the model

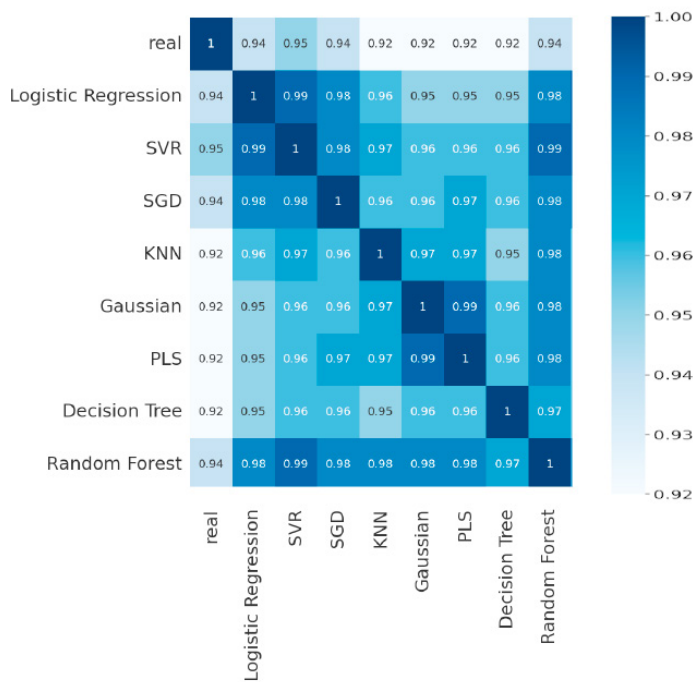


Fig. 7. Correlation between the results created through the different models

4. Conclusion

The best algorithms proved to support vector regression, multilayer perceptron, random forest, and logistic regression. These models showed similarities to each other in the way they produced weather predictions. In terms of each metric, the SVR regressor received the smallest error values and the largest correlation values. For the algorithms

with low error values, the distribution of the predicted data mirrored the characteristics of the real data. Although the created models showed very good performance in tracking atmospheric trends, there were also problems related to the prediction of violent phenomena. This behavior could be reduced by including data from a wider geographic range, which would help analyze the propagation of weather phenomena.

For all models created, daily weather cycles were evident; if a larger time range than the month of January was included, we could also expect to see annual cycles in the weather forecasts created. For the algorithms that achieved larger error values, that is, KNN, decision tree, and Gaussian regression, a poor fit to the real data from the test set can be seen, which is most likely due to over-fitting to the training set. The models in many cases repeated runs that were already observed in the training set. Moreover, an analysis of the distribution of the data created by these models shows a very irregular distribution of observations. In this article, various machine learning models are shown and compared, and those whose application results in the best results are highlighted. Based on the analysis, it is possible to make preliminary recommendations for the implementation of machine learning models for weather forecasting. The results obtained are the baseline for further consideration and provide an opportunity to compare the results obtained.

The results obtained indicate the need for further analysis and deepening of knowledge related to the application of ML in weather forecasting, as well as the development of new algorithms that can cope with the complex nature of atmospheric data. The application of ML-based pure methods has yielded good results and further incorporation into hybrid NWP models is a very promising scope in the development of the field.

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